

Chapter 1 Introduction

1.1 Background and Functional Split Evolution

To address the diverse and stringent requirements of 5G networks, the 3rd Generation Partnership Project (3GPP) explored various functional split options to disaggregate the traditional monolithic base station architecture [?]. Building upon this foundation, the O-RAN Alliance further standardized the disaggregated architecture into three primary logical nodes: the Open Central Unit (O-CU), Open Distributed Unit (O-DU), and Open Radio Unit (O-RU). This disaggregation allows operators to optimize transport bandwidth and computational resources based on their specific deployment environments.



Currently, the industry recognizes three mainstream functional split configurations. The first is **Split Option 2**, which defines the interface between the O-CU and O-DU. The second is **Split Option 7.2**, a low-layer split standardized by O-RAN between the O-DU and O-RU [?]. The third is **Split Option 6**, a MAC-PHY split (often referred to as the O-DU High/Low split), which is the primary focus of this thesis. To ensure interoperability for Option 6, the Small Cell Forum (SCF) has unified the interface specifications under the network Functional Application Platform Interface (nFAPI) [?].

These three mainstream split options exhibit fundamental differences in their synchronization requirements and management mechanisms:

- **Split Option 2 (CU/DU Split):** The split point is located between the PDCP and RLC layers. Because it operates at a higher protocol layer, its synchronization requirement is based on **Sequence Numbers** rather than strict time domains. The core management mechanism relies on **Flow control and Deep Buffering** to handle packet delivery. Consequently, it is well-suited for deployment in **General packet-switched networks** without requiring precise physical-layer timing alignment.
- **Split Option 6 (DU High/Low split):** Operating between the MAC and PHY layers, this split requires a more stringent **Frame/Slot** level of synchronization. Its core management focuses heavily on **Delay management** to ensure scheduling instructions arrive before their intended slot boundaries. Like Option 2, it is designed to operate over **General packet-switched networks**, but it must adaptively adjust to network jitter.
- **Split Option 7.2 (DU/RU Split):** Located within the physical layer, this split demands extreme **Phase/Symbol** level synchronization. Its core management requires strict hardware-assisted mechanisms such as **Hardware PTP and Window** enforcement. Due to its high sensitivity to delay and jitter, it necessitates deployment over specialized **Time-Sensitive Networking (TSN)** environments.

1.2 Comparative Analysis of Timing Maintenance

Table 1.1 summarizes the technical characteristics of the three split architectures regarding synchronization maintenance.

Table 1.1: Comparison of Functional Splits and Timing Maintenance

Split Option	Sync Level	Core Management	Deployment Environment
Option 2 (CU/DU Split)	Sequence Number	Flow control & Deep Buffering	General packet-switched networks
Option 6 (DU High/Low split)	Frame/Slot	Delay management	General packet-switched networks
Option 7.2 (DU/RU Split)	Phase/Symbol	Hardware PTP & Window	Time-Sensitive Networking

This architectural difference means that when facing non-ideal fronthaul networks, Option 6 exhibits a better delay tolerance compared to Option 7.2, while providing more granular physical-layer control compared to Option 2.

1.3 Motivation and Problem Statement

The necessity of investigating high latency in Lower Layer Split (LLS) architectures is highlighted in the O-RAN Fronthaul working group specification [?]. Specifically, Annex L of the specification discusses considerations for operating in non-ideal transport environments. Long latency fron-

thaul links may break traditional Hybrid ARQ (HARQ) timelines, leaving only a tight 4 ms processing budget for the base station. This constraint often necessitates the use of real-time operating systems (RTOS) and dedicated hardware. However, for low- to medium-impact deployment scenarios, such as indoor small cells and fixed wireless access, the performance impact of delayed HARQ or predictive HARQ is negligible [?]. Thus, enforcing sub-millisecond latency constraints restricts the exploration of innovative, cost-effective deployments.

In this context, the nFAPI (Split Option 6) architecture inherently aligns with the requirements of non-ideal fronthaul deployments. By separating the MAC and PHY layers, Split 6 exhibits a much higher latency tolerance (typically 2 ms to 10 ms) compared to Split 7.2x. This enhanced tolerance allows the Virtualized Network Function (VNF) hosting the MAC layer to be deployed on General Purpose Processors (GPPs) in cloud environments, eliminating the strict requirement for expensive dedicated fiber and precise synchronization hardware. Furthermore, nFAPI incorporates timing procedures that enable the MAC layer to adjust its scheduling behavior based on fronthaul latency conditions, directly addressing the issue of broken HARQ windows.

The rationale for migrating the MAC layer to a centralized cloud environment centers on the concept of resource pooling and elasticity. In 5G vRAN architectures, the computational load of the physical layer (L1) is primarily determined by relatively static parameters, such as system bandwidth and antenna configuration. In contrast, the processing complexity of the Layer 2 (L2) stack, particularly the MAC scheduler, exhibits a linear or even quadratic dependency on the number of active User Equipments (UEs)

and traffic volume [?,?]. For instance, as the number of connected UEs increases, the MAC scheduler must continuously evaluate proportional fair metrics, process buffer status reports, and handle logical channels, leading to a substantial surge in CPU cycles [?]. By decoupling the MAC layer into a VNF, the cloud infrastructure can dynamically allocate virtual cores to accommodate fluctuating UE densities, maintaining cost-effective scaling compared to local PNF deployments.

However, this resource scaling introduces a new challenge for synchronization. As the number of UEs increases, the MAC processing time ($T_{\text{process_VNF}}$) inevitably lengthens within a given Transmission Time Interval (TTI). This extended processing duration compresses the remaining time margin available for fronthaul transmission latency. If the scheduling mechanism does not account for this endogenously increasing processing delay, static timing configurations will result in scheduling messages arriving at the PNF too late. Although Option 6 possesses flexible timing adjustment capabilities, its dynamic adjustment algorithms remain immature in existing open-source implementations, such as OpenAirInterface (OAI). Consequently, when confronted with non-deterministic operating system scheduling, UE-dependent processing load, and network congestion, the system frequently encounters “Missing Subframe” errors. This thesis selects Split 6 as the primary research subject, proposing an Adaptive Advance-Time Control architecture to dynamically compensate for both processing and transport latency variations, effectively overcoming these synchronization failures. A detailed analysis of this non-deterministic latency and the corresponding verification of these problems will be presented in Chapter ??, specifically isolating these issues in Scenario I (??).

1.4 Related Works

The evolution of Radio Access Network (RAN) architectures from Distributed RAN (D-RAN) to Open RAN (O-RAN) requires a thorough study of functional splits [?, ?, ?]. The MAC-PHY split (Option 6), standardized by the Small Cell Forum (SCF) as nFAPI [?, ?], is suitable for small cell deployments. It allows the MAC layer to be virtualized on commercial servers while the PHY layer remains on dedicated hardware.

The performance of the nFAPI split depends on the transport network. Practical deployments often use non-ideal fronthaul, such as Shared Ethernet or DOCSIS. Mehran and MacKenzie [?] demonstrated that uplink throughput drops to zero with a small increase in fronthaul delay. Furthermore, server processing delay is unpredictable. Khalaf *et al.* [?] indicated that server delay increases under high CPU utilization, causing the MAC scheduler to miss its timing window.

To address fronthaul latency, Huang and Zhang [?] proposed a distributed MAC scheduling scheme. While this scheme reduces the impact of delayed grants, it introduces a “split-brain” issue with multiple schedulers. Akgun *et al.* [?] utilized intelligent scheduling to adapt modulation based on interference, but did not solve the sub-millisecond timing of message delivery itself. Existing OpenAirInterface (OAI) reference models typically rely on static timing windows, leading to frequent “missing subframe” errors when network jitter exceeds configured limits. This paper bridges this gap by proposing a centralized Adaptive Advance-Time Control algorithm to dynamically manage scheduling timing.

1.5 Contributions

- **Delay Management in nFAPI Split 6:** We analyze and quantify the non-deterministic latency factors, including network jitter and server processing time, that lead to synchronization failures in disaggregated RAN deployments.
- **Adaptive Advance-Time Control Algorithm:** We propose a dynamic timing adjustment algorithm using a dual-loop architecture. The Convergence Optimization Loop estimates end-to-end jitter, while the VNF Timing Actuator adjusts scheduling lead times to ensure punctual delivery of P7 messages.
- **Experimental Validation:** We prototype the proposed mechanism on the OAI platform. Experimental results show that our algorithm eliminates “missing subframe” errors and maintains stable throughput under non-ideal fronthaul environments.

The rest of the thesis is organized as follows. Chapter ?? defines the system model/architecture considered in this paper. The proposed analytical model/method is given in Chapter ??. Chapter ?? shows the numerical results. Concluding remarks are given in Chapter ??.